Do Automated Child Detectors Miss Darker-Skinned Children? A Preregistered Pilot Audit

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Abstract

Text-to-image training data is often filtered with automated child detectors, but it is not known whether these detectors work equally well for children of different skin tones. In this pilot I evaluate five detectors on two face datasets with perceived-race labels and compare true-positive rates (TPR) for White and Black minors after stratifying by age. The analysis plan was preregistered before any detector was run on the full data. On UTKFace, every detector flags Black minors less often than White minors, with gaps of 4.0 to 10.0 percentage points that remain significant after Holm correction. Most of the difference comes from 10–17-year-olds, and the age-estimation models tend to estimate Black minors as older than they are. On FairFace, where only children under ten can be evaluated, I find no difference. Since these are cropped face images, the next phase repeats the analysis on in-the-wild images from ICCWD with skin-tone annotations.

1 Introduction

Crețu et al. [1] evaluated a range of child detectors as a defence against the generation of child sexual abuse material, and found that even the best method misses more than 6% of child images. They also point out that when sub-groups are filtered at different rates, the data that remains is biased. As far as I know, skin tone has not been examined as one of these sub-groups. If detectors miss darker-skinned children more often, a filtered dataset would keep proportionally more images of them, which is the opposite of what the filter is meant to do. This report describes a first, controlled test of that question.

2 Method

I used UTKFace [3], which has 23,705 aligned face images with exact ages, and the validation split of FairFace [4], with 10,954 faces labelled in age bins. Both datasets provide perceived-race labels, which I used as given; the main comparison is between White and Black. Minors (under 18) are used for the TPR and adults aged 20–59 for the false-positive rate (FPR). The FairFace 10–19 bin crosses the age of 18, so I left it out. This gives 1,856 White and 280 Black minors in UTKFace, and 240 and 281 in FairFace.

The five detectors are listed in Table 1. Two estimate age from the face, two are zero-shot CLIP classifiers, and one is a small vision-language model answering a yes/no question. If no face is found, the image counts as not flagged, since a filter would keep it.

Detector	An image is flagged as containing a child if
MiVOLO v2, face only [5]	the youngest detected face is estimated to be under 18, as in [1]
FairFace classifier	the expected age over the nine FairFace age bins is under 18 (UTKFace only)
CLIP ViT-B/32, ViT-L/14 [6]	the best of four child prompts scores higher than the best of three adult prompts
Qwen2-VL-2B [7]	"yes" is more likely than "no" as the answer to "Is there a child (a person under 18 years old) in this image?"

Table 1: Detectors and decision rules.

Older children are harder to detect for every model, so comparing raw rates would penalise whichever group has more teenagers. I therefore computed the TPR within age strata (0–9 and 10–17 for UTKFace, 0–2 and 3–9 for FairFace) and averaged the strata with the same weights for both groups. Confidence intervals come from 2,000 bootstrap resamples within each group, p-values from a permutation test that shuffles the group labels within strata, and Holm's method corrects for testing several detectors on the same dataset. I also looked at face-detection failures, errors in the estimated age, false positives on adults, a logistic regression adjusting for age and image quality, and per-group decision thresholds.

The hypotheses and analysis were written down and frozen before the full runs. Three things changed afterwards, and all are recorded in the preregistration. The original FairFace model weights were no longer available, so I used a FairFace-trained ViT with the same age bins. After seeing the results, I replaced the

bootstrap intervals for face-detection failure with Wilson intervals, because the bootstrap gives a zero-width interval when a group has no failures. I also found that the CLIP models had first been run with the wrong activation function (GELU instead of the QuickGELU they were trained with); I fixed this and reran them. All results below use the corrected models, and no conclusion changed.

3 Results

Dataset	Detector	TPR White	TPR Black	Gap	95% CI	Holm p
UTKFace	MiVOLO v2	94.0	85.6	8.4	5.3 to 11.9	0.002
	FairFace classifier	86.4	82.4	4.0	0.4 to 7.8	0.016
	CLIP ViT-B/32	97.9	90.4	7.5	4.5 to 10.6	0.002
	CLIP ViT-L/14	95.1	85.1	10.0	6.9 to 13.3	0.002
	Qwen2-VL-2B*	92.1	86.0	6.1	2.0 to 10.3	0.014
FairFace	MiVOLO v2	96.3	95.9	0.4	−3.1 to 3.5	1.000
	CLIP ViT-B/32	99.1	99.0	0.1	−1.5 to 1.7	1.000
	CLIP ViT-L/14	98.2	99.3	−1.1	−3.1 to 0.7	1.000
	Qwen2-VL-2B*	95.2	97.9	−2.7	−7.1 to 1.7	1.000

Table 2: Age-stratified TPR (%) and the White − Black gap in percentage points. *Qwen2-VL was run on a preregistered stratified subsample (1,844 UTKFace and 1,949 FairFace images).

On UTKFace, all five detectors show a positive gap whose confidence interval excludes zero, so the preregistered hypothesis holds there (Table 2, Figure 1). On FairFace, none of the four detectors shows a gap.

On UTKFace the difference comes almost entirely from teenagers. Among 10–17-year-olds, MiVOLO flags 81% of White minors and 56% of Black minors; for the two CLIP models and Qwen2-VL the figures are 94% and 74%, 83% and 55%, and 80% and 58%. Under ten, the gap is at most 2.8 points for any detector. FairFace only allows children under ten to be tested, which is probably why it shows no gap, although this explanation was not part of the original plan.

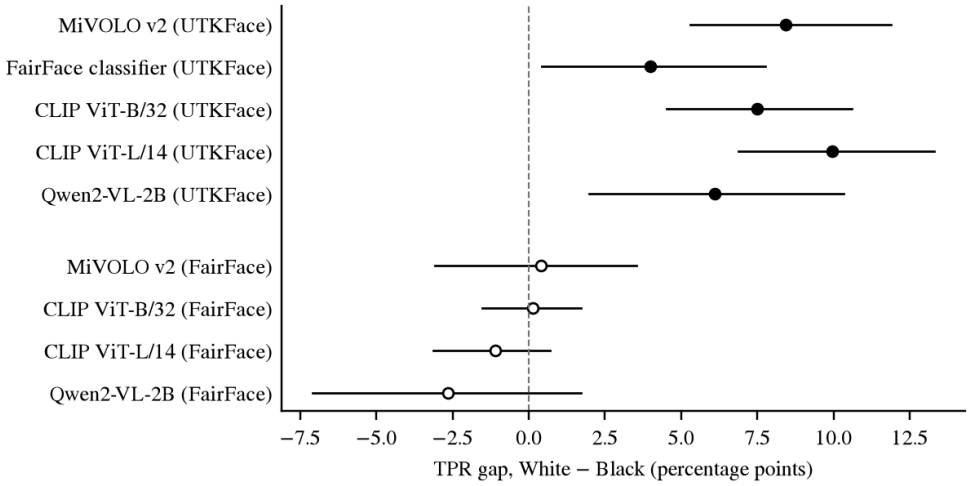


Figure 1: White − Black TPR gap with 95% bootstrap confidence intervals. Filled markers: UTKFace; open markers: FairFace.

The age-estimation models suggest one reason. Among minors whose face was found, MiVOLO's age estimates for Black minors are on average 1.52 years higher, relative to their true age, than for White minors on UTKFace (95% CI 0.95 to 2.28), and 1.22 years higher on FairFace (0.83 to 1.63, using bin midpoints). The FairFace classifier shows a smaller difference of 0.72 years. Face detection does not explain the gap: on UTKFace, MiVOLO missed the face in 0 of the 280 Black minors' images and in 12 of the 1,856 White minors' images. The gap also remains in a logistic regression that adjusts for age, brightness, blur and face size, with odds ratios for Black minors of 0.32 for MiVOLO, 0.25 and 0.22 for the CLIP models, and 0.53 for Qwen2-VL. For the FairFace classifier the odds ratio is 0.81 and not significant.

Lowering the decision threshold for Black minors until their TPR matches the overall rate is possible, but costly. The share of Black adults flagged as children rises from 0.5% to 7.4% for MiVOLO, and from 7.4% to 58.3% for CLIP ViT-B/32.

4 Limitations

This is a pilot and has clear limits. The images are cropped and aligned faces, which are much easier than real training data; [1] reports that MiVOLO detects only 46–59% of children in the wild. The groups are perceived-race labels rather than measured skin tone. UTKFace contains only 280 Black minors, 126 of them teenagers, and its age and race labels are known to be noisy. Finally, my detectors are not identical to those in [1]: the MiVOLO version used there is not stated, and my FairFace model is a substitute with a different rule for the 10–19 age bin.

5 Next steps

The next phase uses ICCWD [2], which contains 10,000 human-labelled image–caption pairs from CC3M, 1,675 of them with children, and is the benchmark used in [1]. I plan to annotate the child images and a sample of adult images with the Monk Skin Tone scale, using two annotators, and then rerun the same detectors together with caption-based and combined image-and-caption detectors. This will show whether the gap holds on realistic images and whether combining the two modalities reduces it. As in this phase, the analysis will be preregistered before any detector is run.

References

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